

Tables

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Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_TAB_001	55	—	$\begin{tabular}[t]{ l l } \hline \multicolumn{2}{ l }{Gravitational gauge field variables} \\ \hline \\\hline & \begin{tabular}[t]{ l l } \hline \\\hline (h_{\mu}) & \text{translational gauge field (}h\text{-field)} \\ \hline (h_{\mu}) & \text{determinant of }h_a{}^{\mu} \\ \hline (b^a{}_{\mu}) & \text{inverse }h\text{-field} \\ \hline (A^{ab}{}_{\mu}) & \text{rotational gauge field (}A\text{-field)} \\ \hline (B_{\mu}) & \text{WGT dilational gauge field (}B\text{-field)} \\ \hline (V_{\mu}) & \text{eWGT dilational gauge field (}V\text{-field)} \\ \hline A^{\dagger ab}{}_{\mu} \equiv A^{ab}{}_{\mu} + 2\eta^{c[ab]}{}_{\mu} h_c{}^{\nu} V_{\nu} & \text{eWGT extended rotational gauge field (}A^{\dagger}\text{-field)} \\ \hline \end{tabular} \\ \hline \end{tabular}$ Derivative operators $\begin{aligned} D_a &\equiv h_a{}^{\mu} D_{\mu} \equiv h_a{}^{\mu} (\partial_{\mu} + \tfrac{1}{2} A^{ab}{}_{\mu} \Sigma_{ab}) \\ \partial^a_{\mu} &\equiv \partial_{\mu} + w B_{\mu} \\ D^a_{\mu} &\equiv h_a{}^{\nu} D^a_{\nu} \equiv h_a{}^{\nu} (\partial^a_{\nu} + \tfrac{1}{2} A^{ab}{}_{\nu} \Sigma_{ab}) \\ \partial^{\dagger}_{\mu} &\equiv \partial_{\mu} - w (V_{\mu} + \tfrac{1}{3} T_{\mu}) \\ D^{\dagger}_{\mu} &\equiv h_a{}^{\nu} D^{\dagger}_{\nu} \equiv h_a{}^{\nu} (\partial^{\dagger}_{\nu} + \tfrac{1}{2} A^{\dagger ab}{}_{\nu} \Sigma_{ab}) \\ D^2_a &\equiv h_a{}^{\mu} D^2_{\mu} \equiv h_a{}^{\mu} (\partial_{\mu} + \tfrac{1}{2} A^{\dagger ab}{}_{\mu} \Sigma_{ab}) \end{aligned}$ Gauge field strengths $\begin{aligned} \mathcal{R}^{ab}{}_{cd} &= \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]}) \\ \mathcal{R}^{\dagger ab}{}_{cd} &= 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]}) \\ T^{ab}{}_{bc} &= 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu} b^a{}_{\nu]} \\ T^{*ab}{}_{bc} &= 2h_b{}^{\mu} h_c{}^{\nu} D^a_{[\mu} b^a{}_{\nu]} \\ T^{[a}{}_{bc} &= 2h_b{}^{\mu} h_c{}^{\nu} D^a_{[\mu} b^a{}_{\nu]} \\ T^{3a}{}_{bc} &= 2h_b{}^{\mu} h_c{}^{\nu} D^2_{[\mu} b^a{}_{\nu]} \\ \mathcal{H}_{ab} &= 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} B_{\nu]} \\ \mathcal{H}^{\dagger}_{ab} &= 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} (V_{\nu} + \tfrac{1}{3} T_{\nu]}) \end{aligned}$ Reduced A-fields and related quantities $\begin{aligned} c^{ab}{}_{bc} &\equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu} b^a{}_{\nu]} \\ c^{*ab}{}_{bc} &\equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial^a_{[\mu} b^a{}_{\nu]} \\ c^{\dagger ab}{}_{bc} &\equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial^{\dagger}_{[\mu} b^a{}_{\nu]} \\ {}^0A^{ab}{}_{\mu} &\equiv \tfrac{1}{2} b^c{}_{\mu} (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab}) \\ {}^0A^{*ab}{}_{\mu} &\equiv \tfrac{1}{2} b^c{}_{\mu} (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*ab}) \\ {}^0A^{\dagger ab}{}_{\mu} &\equiv \tfrac{1}{2} b^c{}_{\mu} (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c{}^{\dagger ab}) \end{aligned}$ Currents and related quantities $\begin{aligned} \mathcal{L} &\equiv h^{-1} L \\ t^a{}_b &\equiv t^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_G / \delta h_a{}^{\mu} \\ s_{ab}{}^c &\equiv s_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_G / \delta A^{ab}{}_{\mu} \\ j^a &\equiv j^a{} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_G / \delta B_{\mu} \\ t^{\dagger a}{}_b &\equiv t^a{}_b + 2(s_{cb}{}^a V^c - s^{ac}{}_c V_b) \\ j^{\dagger a} &\equiv j^a - 2s^{ab}{}_b \\ \tau^a{}_b &\equiv \tau^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_M / \delta h_a{}^{\mu} \\ \sigma_{ab}{}^c &\equiv \sigma_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_M / \delta A^{ab}{}_{\mu} \\ \zeta^a &\equiv \zeta^a{} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_M / \delta B_{\mu} \\ \tau^{\dagger a}{}_b &\equiv \tau^a{}_b + 2(\sigma_{cb}{}^a V^c - \sigma^{ac}{}_c V_b) \\ \zeta^{\dagger a} &\equiv \zeta^a - 2s^{ab}{}_b \end{aligned}$ <		

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1510.06699_TAB_004	55	—	<pre>\begin{tabular}[t]{l}{l} \hline \(\mathcal{R}^{\wedge\{a\}b\{ }_{\{c\}d\}}=\mathcal{R}^{\wedge\{*a\}b\{ }_{\{c\}d}=2\,h_{\{a\}\{ }^{\wedge\{\mu\}}h_{\{b\}\{ }^{\wedge\{\nu\}}\left(\partial_{[\mu}\mathcal{A}^{\wedge\{a\}b\{ }_{\{c\}\{\nu\}}]+A^{\wedge\{a\}c\{b\}^{\nu\}}_{[\mu}\mathcal{A}^{\wedge\{c\}b\{ }_{\{c\}\{\nu\}}\right)\) \mathcal{R}^{\wedge\{\dagger a\}b\{ }_{\{c\}d}=2\,h_{\{a\}\{ }^{\wedge\{\mu\}}h_{\{b\}\{ }^{\wedge\{\nu\}}\mathcal{D}^{\dagger}_{[\mu}\mathcal{A}^{\wedge\{\dagger a\}b\{ }_{\{c\}\{\nu\}}]+A^{\dagger\{a\}c\{b\}^{\nu\}}_{[\mu}\mathcal{A}^{\wedge\{\dagger c\}b\{ }_{\{c\}\{\nu\}}\right) \mathcal{T}^a{}_{bc}=2\,h_b{}^\mu h_c{}^\nu D_{[\mu}b^a{}_{\nu]} \mathcal{T}^{*a}{}_{bc}=2\,h_b{}^\mu h_c{}^\nu D^*_{[\mu}b^a{}_{\nu]} \mathcal{T}^{\dagger a}{}_{bc}=2\,h_b{}^\mu h_c{}^\nu D^{\dagger}_{[\mu}b^a{}_{\nu]} \mathcal{T}^{\natural a}{}_{bc}=2\,h_b{}^\mu h_c{}^\nu D^{\natural}_{[\mu}b^a{}_{\nu]} \mathcal{H}_{ab}=2\,h_a{}^\mu h_b{}^\nu \partial_{[\mu}B_{\nu]} \mathcal{H}^{\dagger}_{ab}=2\,h_a{}^\mu h_b{}^\nu \partial_{[\mu}(V_{\nu]}+\tfrac{1}{3}T_{\nu]}) \end{tabular}</pre>	$\mathcal{R}^{ab}_{cd} = \mathcal{R}^{*ab}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}_{\nu]} + A^a{}_{c[\mu} A^{cb}_{\nu]})$$\mathcal{R}^{\dagger ab}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}_{\nu]})$$\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$$\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu} b^a{}_{\nu]}$$\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^{\dagger}_{[\mu} b^a{}_{\nu]}$$\mathcal{T}^{\natural a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D^{\natural}_{[\mu} b^a{}_{\nu]}$$\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$$\mathcal{H}^{\dagger}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_{\nu]} + \tfrac{1}{3} T_{\nu]})$	
1510.06699_TAB_005	55	—	<pre>\begin{tabular}[t]{l}{l} \hline \((c^{\wedge\{a\}\{ }_{\{b\}c}) \equiv 2\,h_{\{b\}\{ }^{\wedge\{\mu\}}h_{\{c\}\{ }^{\wedge\{\nu\}}\partial_{[\mu}b^{\wedge\{a\}\{ }_{\{b\}c\}}]_{\{\nu\}}\) \\ \hline \((c^{\wedge\{*a\}\{ }_{\{b\}c}) \equiv 2\,h_{\{b\}\{ }^{\wedge\{\mu\}}h_{\{c\}\{ }^{\wedge\{\nu\}}\partial_{[\mu}b^{\wedge\{*a\}\{ }_{\{b\}c\}}]_{\{\nu\}}\) \\ \hline \((c^{\wedge\{\dagger a\}\{ }_{\{b\}c}) \equiv 2\,h_{\{b\}\{ }^{\wedge\{\mu\}}h_{\{c\}\{ }^{\wedge\{\nu\}}\mathcal{D}^{\dagger}_{[\mu}b^{\wedge\{\dagger a\}\{ }_{\{b\}c\}}]_{\{\nu\}}\) \\ \hline \((c^{\wedge\{\natural a\}\{ }_{\{b\}c}) \equiv 2\,h_{\{b\}\{ }^{\wedge\{\mu\}}h_{\{c\}\{ }^{\wedge\{\nu\}}\mathcal{D}^{\natural}_{[\mu}b^{\wedge\{\natural a\}\{ }_{\{b\}c\}}]_{\{\nu\}}\) \\ \hline \((\{ }^{\wedge\{0\}}A^{\wedge\{a\}b\{ }_{\{c\}\{\mu\}}) \equiv \frac{1}{2}b^c{}_{\mu}(c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})\) \\ \hline \((\{ }^{\wedge\{0\}}A^{*ab}{}_{\mu} \equiv \frac{1}{2}b^c{}_{\mu}(c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*ab})\) \\ \hline \((\{ }^{\wedge\{0\}}A^{\dagger ab}{}_{\mu} \equiv \frac{1}{2}b^c{}_{\mu}(c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})\) \end{tabular}</pre>	$c^a{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$$c^{*a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial^*_{[\mu} b^a{}_{\nu]}$$c^{\dagger a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \mathcal{D}^{\dagger}_{[\mu} b^a{}_{\nu]}$$c^{\natural a}{}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \mathcal{D}^{\natural}_{[\mu} b^a{}_{\nu]}$${}^0A^{ab}{}_{\mu} \equiv \tfrac{1}{2}b^c{}_{\mu} (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$${}^0A^{*ab}{}_{\mu} \equiv \tfrac{1}{2}b^c{}_{\mu} (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*ab})$${}^0A^{\dagger ab}{}_{\mu} \equiv \tfrac{1}{2}b^c{}_{\mu} (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$	

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1510.06699_TAB_006	55	—	$\begin{tabular}[t]{ l } \hline \mathcal{L} \equiv h^{-1} L \\ \equiv t^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_G / \delta h_a{}^{\mu} \\ s_{ab}{}^c \equiv s_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_G / \delta A^{ab}{}_{\mu} \\ j^a \equiv j^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_G / \delta B_{\mu} \\ t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b) \\ j^{\dagger a} \equiv j^a - 2s^{ab}{}_b \\ \tau^a{}_b \equiv \tau^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_M / \delta h_a{}^{\mu} \\ \sigma_{ab}{}^c \equiv \sigma_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_M / \delta A^{ab}{}_{\mu} \\ \zeta^a \equiv \zeta^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_M / \delta B_{\mu} \\ \tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b) \\ \zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b \end{tabular}$	<table><tr><td>L</td></tr><tr><td>$\mathcal{L} \equiv h^{-1} L$</td></tr><tr><td>$t^a{}_b \equiv t^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_G / \delta h_a{}^{\mu}$</td></tr><tr><td>$s_{ab}{}^c \equiv s_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_G / \delta A^{ab}{}_{\mu}$</td></tr><tr><td>$j^a \equiv j^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_G / \delta B_{\mu}$</td></tr><tr><td>$t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$</td></tr><tr><td>$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$</td></tr><tr><td>$\tau^a{}_b \equiv \tau^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_M / \delta h_a{}^{\mu}$</td></tr><tr><td>$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_M / \delta A^{ab}{}_{\mu}$</td></tr><tr><td>$\zeta^a \equiv \zeta^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_M / \delta B_{\mu}$</td></tr><tr><td>$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$</td></tr><tr><td>$\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$</td></tr></table>	L	$\mathcal{L} \equiv h^{-1} L$	$t^a{}_b \equiv t^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_G / \delta h_a{}^{\mu}$	$s_{ab}{}^c \equiv s_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_G / \delta A^{ab}{}_{\mu}$	$j^a \equiv j^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_G / \delta B_{\mu}$	$t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$	$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	$\tau^a{}_b \equiv \tau^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_M / \delta h_a{}^{\mu}$	$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_M / \delta A^{ab}{}_{\mu}$	$\zeta^a \equiv \zeta^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_M / \delta B_{\mu}$	$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$	$\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$	$\begin{aligned} L \\ \mathcal{L} &\equiv h^{-1} L \\ t^a{}_b &\equiv t^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_G / \delta h_a{}^{\mu} \\ s_{ab}{}^c &\equiv s_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_G / \delta A^{ab}{}_{\mu} \\ j^a &\equiv j^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_G / \delta B_{\mu} \\ t^{\dagger a}{}_b &\equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b) \\ j^{\dagger a} &\equiv j^a - 2s^{ab}{}_b \\ \tau^a{}_b &\equiv \tau^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_M / \delta h_a{}^{\mu} \\ \sigma_{ab}{}^c &\equiv \sigma_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_M / \delta A^{ab}{}_{\mu} \\ \zeta^a &\equiv \zeta^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_M / \delta B_{\mu} \\ \tau^{\dagger a}{}_b &\equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b) \\ \zeta^{\dagger a} &\equiv \zeta^a - 2s^{ab}{}_b \end{aligned}$
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